



Sam2Rad: A segmentation model for medical images with learnable prompts

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ABSTRACT

The Segment Anything Model and its variants, such as MedSAM, have demonstrated potential for medical image segmentation. However, they heavily rely on high-quality manual prompts, which are both time-consuming and require domain expertise. Even when using manual prompts, including sparse prompts like boxes, points, or text, and dense prompts such as masks, SAM and its variants like MedSAM (fine-tuned on medical images) fail to segment bones in ultrasound images due to significant domain shift.

To address these limitations, we propose **Sam2** for **Radiology** (Sam2Rad), a framework that extends SAM and its recent iteration SAM2 to segment bony regions in ultrasound images without requiring manual prompts. At the core of our approach is a Prompt Predictor Network (PPN) that uses a lightweight cross-attention mechanism to generate bounding box coordinates, mask prompts, and high-dimensional embeddings to be used as prompts. Specifically, PPN leverages hierarchical feature maps extracted from SAM's image encoder as keys and values, and learnable object embeddings as queries. The output of the cross-attention is then used to predict the bounding box coordinates, mask prompts, and high-dimensional prompts. These predicted prompts are subsequently fed to SAM's mask decoder to generate the final segmentation mask. By aligning the learned prompts with SAM's original training scheme, PPN enhances compatibility with SAM's architecture, requiring no additional standalone encoders.

To preserve SAM's extensive world knowledge, we keep all SAM modules frozen and train PPN only. This approach enables efficient parameter utilization while retaining SAM's generalization capabilities. Additionally, Sam2Rad can operate in three modes: fully autonomous without human supervision, semi-autonomous with *human-in-the-loop* for iterative refinement, and fully manual for tasks like data labeling.

We tested the proposed model – Sam2Rad on 3 musculoskeletal US datasets – wrist (3822 images), shoulder rotator cuff (1605 images), and hip (4849 images). Without Sam2Rad, all SAM2 variants failed to segment shoulder US in zero-shot generalization with bounding box prompts. Our model, Sam2Rad, improved the performance of all SAM base networks in all datasets, without requiring manual prompts. The improvement in dice score ranged from a 2.2%–5.8% for hip, 19.6%–32.8% for wrist, and up to 51.3% improvement in Dice score (from 25.2% to 76.5% Sam2 large) on shoulder data. Notably, Sam2Rad could be trained with as few as 10 labeled images and it is compatible with any SAM architecture.

The code is available at <https://github.com/aswahd/SamRadiology>.

1. Introduction

Automated segmentation of structures in medical images is an essential step for various computer-aided diagnoses and treatment planning procedures. Over the past decade, the evolution of deep learning models has significantly accelerated the progress of automated medical image segmentation.

Traditionally, segmentation tasks are approached separately by specialized networks UNet [1], Mask R-CNN [2], nnUNet [3], UNet++ [4],

Attention UNet [5], TransUNet [6], UNETR [7], Swin-UNet [8]. As a result, their use in broader clinical contexts is limited as this approach requires substantial amounts of data to train each task and ignores correlations between different segmentation objectives.

Deep learning scaling laws have demonstrated that increasing both the size of training datasets and the number of model parameters significantly enhances model performance [9,10]. This insight has led to the development of foundation models—large, pre-trained neural networks capable of performing diverse tasks with minimal fine-tuning.

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Foundation models, such as OpenAI's CLIP for vision and language tasks [11] and SAM for segmentation [12], encapsulate broad world knowledge, enabling them to generalize across various downstream tasks. Their adaptability and robustness make them particularly promising for domains like medical imaging, where labeled data is scarce and task variability is high.

Foundation models, trained on extensive datasets encompassing diverse information, have demonstrated impressive capabilities in generalizing to downstream tasks with minimal labeled data [13–15]. They are robust to adversarial attacks [16], resilient to domain shifts [17,17], and effective in zero-shot settings [11,12,18–21]. These characteristics have driven their popularity in medical imaging, where labeled data is limited and diverse. However, significant challenges remain, particularly when adapting foundation models to complex medical imaging modalities like ultrasound. These models often fail to account for the unique imaging artifacts and high variability inherent in medical data. Addressing these challenges requires innovative strategies to bridge the gap between general-purpose foundation models and the specific demands of medical applications.

Segment Anything Model (SAM) [12,19] is a promptable foundation model that has demonstrated impressive zero-shot generalization to various medical imaging modalities, making it a versatile tool for segmentation tasks [22–24]. SAM requires user-provided prompts, such as bounding boxes, points, or masks, to generate segmentation predictions. This prompt-based mechanism allows SAM to adapt to diverse applications. However, it introduces a dependency on high-quality manual input, which can be time-consuming and requires domain expertise.

Even variants fine-tuned on medical images, such as MedSAM [25], still rely on prompts, reducing their suitability for fully automated workflows. SAM's grid-based prompting approach, where a regular grid of points (e.g., 32×32) are sampled across the image, and a non-maximum suppression (NMS) is then applied to filter the predictions, is aimed at automated workflows. While this method eliminates the need for user-provided prompts, it performs poorly on medical images—where point-based prompts struggle to capture the contextual information needed to differentiate between foreground and background regions [26]. Similarly, object detectors such as Grounding DINO [27], have been employed to predict mask or bounding box prompts (GroundedSAM [28]). These models are capable of generating prompts for SAM by detecting objects within the image. However, this approach introduces significant computational overhead, as these models require additional standalone image encoders. Moreover, object detectors typically struggle in low-data scenarios, such as when only a few labeled images are available, limiting their effectiveness in medical imaging applications where annotated datasets are scarce.

In contrast, our hypothesis is that bounding boxes or masks can be directly predicted using SAM's pretrained image encoder, eliminating the need for a separate detection model. By leveraging the hierarchical feature maps already extracted by SAM's encoder, we aim to develop a lightweight and efficient method for prompt prediction. This approach not only reduces computational costs but also ensures compatibility with SAM's architecture, making it more suitable for the low-data regime commonly encountered in medical imaging tasks. Building on this idea, we present **Sam2** for **Radiology** (**Sam2Rad**), which introduces a Prompt Predictor Network (PPN) to generate high-quality prompts automatically. PPN uses a lightweight cross-attention mechanism to produce bounding box coordinates, mask prompts, and high-dimensional embeddings ($N \times 256$) from SAM's encoder features. These are then fed into SAM's mask decoder, enabling fully automated segmentation without reliance on manual or grid-based prompting.

2. Related works

2.1. Foundation models

Foundation models, such as GPT for natural language processing [13], CLIP for vision-and-language tasks [11], and SAM for image segmentation [12], are large-scale deep learning models trained on extensive datasets. These models generalize effectively across diverse downstream tasks, with CLIP excelling at jointly learning visual and textual representations and SAM leveraging over 1 billion annotated masks to establish itself as a robust segmentation foundation model. SAM's ability to handle a variety of segmentation prompts—points, bounding boxes, or masks inspired a wave of research into universal medical image segmentation models [25]. SAM's latest iteration, SAM2 [19], extends its utility to both image and video segmentation, integrating a memory bank to store image/prompt history, and a memory attention mechanism that allows frame-to-frame consistency in video tasks to model object permanence. Notably, SAM2 achieves $6\times$ speed improvements over its predecessor, making it highly efficient for real-time applications. These advancements have established SAM and SAM2 as benchmarks for segmentation tasks, including in medical imaging, where they address challenges like limited labeled data and domain-specific generalization.

However, studies that explored SAM's application in the medical image domain have identified its limitations in some medical modalities [22,26,34–36]. Efforts have been made to address this by fine-tuning SAM on large curated datasets in medical domain, resulting in medical foundation models like MedSAM [25] and SAM-Med2D [37]. Nonetheless, these models' performance remains limited to the modalities they were trained on and heavily relies on prompt quality. This poses a significant challenge when these models are applied to substantially different data distributions such as ultrasound. As shown in Fig. 2, SAM and its variants have sub-optimal performance on US data. This could be due to the unique nature of imaging artifacts seen in US, like the speckle noise, which occurs when the distance between adjacent scatters is less than the cell resolution. Addressing such limitations would require domain-specific fine-tuning for optimal performance while maintaining broad applicability. Furthermore, these foundation models require manual prompts, which are practically difficult without the necessary medical expertise. As a result, deep learning methods for automatically generating appropriate prompts have become a significant focus of research [28–30,32,33,38,39].

2.2. Prompt learning

Traditional approaches to training deep learning models often involve either training the model from scratch or pretraining it with unlabeled data followed by fine-tuning with labeled data for specific tasks. In contrast, prompt-based foundation models can automatically adapt to various tasks by utilizing cues provided by users. These models leverage the domain knowledge introduced through prompts to guide their performance, enabling more efficient and flexible task adaptation without extensive task-specific training.

SAM demonstrates impressive zero-shot generalization to natural images. However, it requires prompts – such as points, boxes, text, or masks – to segment objects in an image, even during inference. Prompts can be either provided by humans (e.g., drawing boxes around objects or inputting text) or predicted by standalone models. Grounded SAM [28] uses Grounding DINO [27] to generate the bounding boxes for all objects in a given image, which are then used to prompt SAM. The limitation of this approach is that it requires a model with a separate image encoder trained from scratch.

Recently, automatic techniques like PerSAM [29] and Matcher [30] have been proposed to generate bounding box prompts for natural images. These approaches assume that foreground and background patches are dissimilar and use patch similarities between a reference image and a test image to generate prompts. However, in US images foreground and background patches are often hard to distinguish,

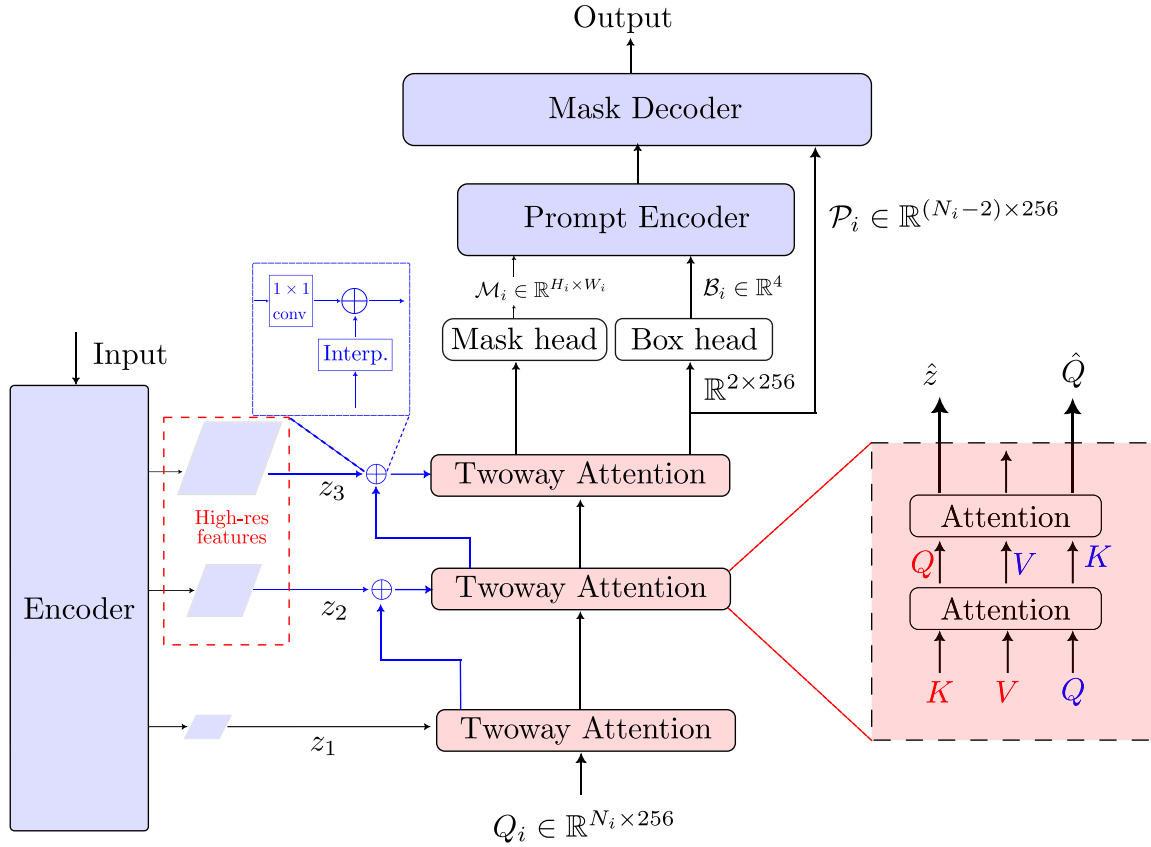


Fig. 1. Sam2Rad architecture The Sam2Rad architecture incorporates a (hierarchical) two-way attention module to predict prompts for queried objects. Each object/class i is represented by learnable queries $Q_i \in \mathbb{R}^{N_i \times 256}$. The Prompt Predictor Network (PPN) predicts bounding box coordinates of the target object (B_i), an intermediate mask prompt ($\mathcal{M}_i$), and $\mathbb{R}^{N_i \times 256}$ high-dimensional prompt embeddings ($\mathcal{P}_i$). The prompt embeddings ($\mathcal{P}_i$) can represent various prompts suitable for the task, such as several point prompts or high-level semantic information. The predicted prompts (i.e., $\mathcal{M}_i$, B_i , & $\mathcal{P}_i$) are then fed to SAM's mask decoder to generate the final segmentation mask. PPN also supports multi-class segmentation by using class-specific queries (Q_i).

resulting in a high cosine similarity between unrelated patches. To address these limitations, we propose a Prompt Predictor Network (PPN) that leverages features directly extracted from SAM's pretrained image encoder. Unlike methods that depend on standalone models like Grounding DINO [27], PPN predicts precise prompts for regions of interest through a lightweight cross-attention mechanism. This mechanism uses hierarchical feature maps from SAM's encoder as keys and values, and learnable embeddings as queries. By dynamically attending to the most relevant features, PPN learns a non-linear function to predict the location of the target regions in the image. The details of the proposed prompt prediction network are discussed in Section 3.1.1.

To the best of our knowledge, this is the first work to report prompt learning for SAM2 [19], the latest version of SAM. For a comparison between our prompt predictor network and other prompting techniques, see Fig. 3.

2.3. Parameter-efficient fine-tuning

Several studies have shown that SAM fails to generalize to medical images even with manual prompts [22,26,34–36]. We observed similar limitations with ultrasound (US) images, as illustrated in Fig. 2.

A straightforward approach to address this issue is to fine-tune all the model parameters (i.e., vanilla fine-tuning). In this method, the entire network is updated using labeled data from the target domain. However, vanilla fine-tuning is often suboptimal as it requires substantial amounts of labeled data, is computationally expensive, and can result in overfitting or performance degradation when labeled data is scarce. For example, CoOp [40] reported a 40% drop in classification accuracy when vanilla fine-tuning the CLIP image encoder. Parameter-efficient fine-tuning (PEFT) methods have emerged as an efficient

alternative to adapt foundation models to specific tasks with minimal modifications to the original model. PEFT methods like adapters [41] and low-rank adaptation (LoRA) [42] are designed to adapt foundation models to downstream tasks with minimal changes to the model's parameters. PEFT methods are particularly useful when labeled data is limited. Adapters [41] insert small modules in parallel to the transformer layers of a foundation model and fine-tune only these new parameters. LoRA [43] is another PEFT method that fine-tunes only a small subset of the model's parameters. Instead of directly updating the pretrained model weight matrices, LoRA learns low-rank matrices that approximate the weight changes needed for the new task. Studies show integrating PEFT with SAM not only reduces computational costs but also mitigates overfitting and improves the overall segmentation accuracy [44].

While the Prompt Predictor Network (PPN) generates high-quality prompts, there are scenarios where a frozen SAM, even when provided with optimal prompts, fails to produce accurate segmentations. These limitations often arise from the inherent challenges of certain datasets, such as extreme variability or unique imaging artifacts, as seen in the shoulder dataset (Table 3). In such instances, we address this issue by fine-tuning the mask decoder using Low-Rank Adaptation (LoRA). By focusing solely on fine-tuning the mask decoder, we preserve the extensive knowledge embedded in SAM's image encoder while enhancing segmentation performance for challenging tasks.

2.4. Contributions

We propose a prompt prediction network that directly predicts N 256-dimensional embeddings alongside bounding box coordinates and

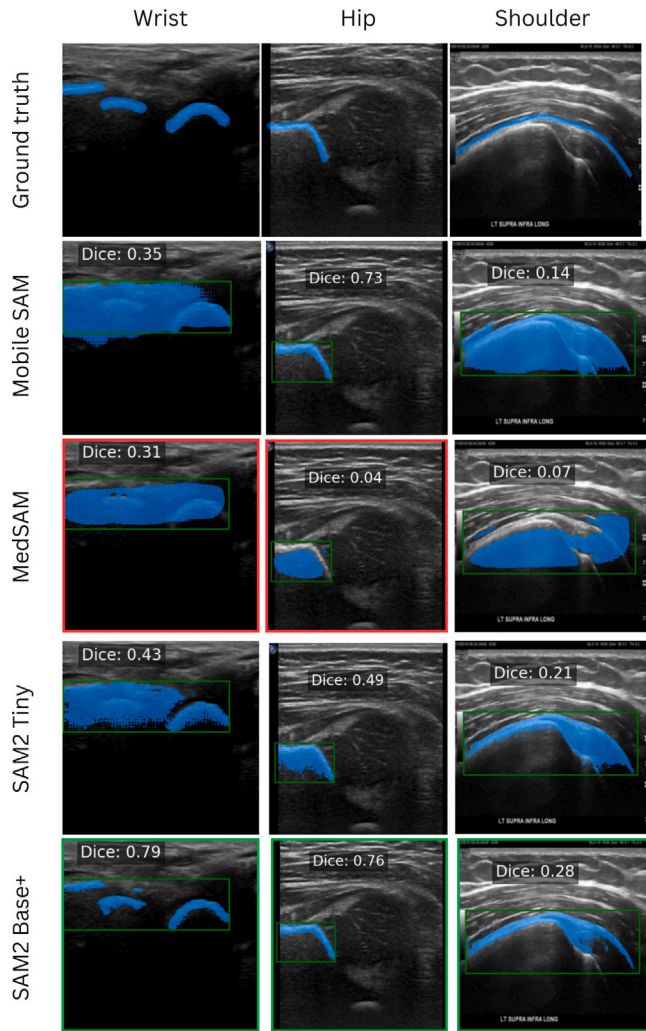


Fig. 2. SAM variants show clear limitations when segmenting US images using bounding box prompts. Surprisingly, MedSAM [25], despite being trained on a large-scale medical dataset, performs the worst. In contrast, SAM2 [19], SAM's latest iteration, demonstrates the best zero-shot generalization. However, smaller SAM2 variants (such as Sam2-Tiny) still incorrectly include background in the segmentation mask. Notably, all models struggle with shoulder US. These observations suggest that bounding box prompts may not be sufficient, or that the models require fine-tuning. In Section 3.1.1, we demonstrate that our proposed prompt learning approach can improve the performance of these models while keeping their pretrained parameters frozen.

mask prompts. The box and mask prompts align the prompt generator to SAM's prompt encoder and the high-dimensional prompts allows the network to learn an abstract representation of a prompt instead of a geometric representation like a box. Moreover, N can be set to any value, in contrast to other methods that typically predict a single bounding box per object. Studies show that adapting SAM to medical imaging often requires several-point prompts. In PPN, N can represent any number of points or other object-relevant information. For instance, for objects that are challenging to segment, such as thin overlapping tissues, the network can adapt by predicting multiple point prompts or using alternative representations.

To the best of our knowledge, this is the first work to report prompt learning for SAM2 [19], the latest version of SAM.

The code is available on <https://github.com/aswahd/SamRadiology>.

3. Methodology

We propose a novel framework, **Sam2 for Radiology (Sam2Rad)**, designed for segmenting medical images without the need for manual prompts. This paper focuses specifically on applying this framework to ultrasound imaging. SAM2Rad incorporates a prompt predictor (generator) network (PPN) that predicts candidate bounding box proposals, mask prompts, and N 256-dimensional prompts. Similar to object detection models such as Mask R-CNN [2], where a bounding box regression head use features from the backbone network to predict coordinates, PPN's bounding box regression head can be viewed as object detection. It utilizes features from SAM's image encoder to generate appropriate prompts for segmenting target regions, functioning analogously to Mask R-CNN's region proposal network (RPN) [2].

3.1. Sam2Rad

The proposed SAM2Rad framework comprises a pretrained SAM/SAM2 encoder, mask decoder, prompt encoder, and prompt predictor network (Fig. 1).

Image encoder. The image encoder extracts features from the input image. It is based on the vision transformer (ViT) architecture [45] and pretrained using masked autoencoders (MAE) [46]. For an input image of size $3 \times 1024 \times 1024$, the image encoder outputs embeddings of size $256 \times 64 \times 64$.

Mask decoder. The mask decoder transforms the image embeddings and prompt embeddings to a final mask.

Prompt encoder. The prompt encoder processes both sparse prompts (points and bounding boxes) and dense prompts (masks). Points are encoded into 256-dimensional embeddings, combining positional and learned embeddings. Bounding boxes are similarly encoded, using positional encodings for the top-left corner and learned embeddings for the bottom-right corner. Masks (256×256) are transformed into $256 \times 64 \times 64$ embeddings using a $4\times$ downsampling convolution block.

To align the learned prompts with the manual prompts SAM was trained on, our predictor network outputs bounding box coordinates, mask prompts, and N 256-dimensional embeddings. For a comparison between our prompt predictor network and other prompting techniques, see Fig. 3. In the next section, we outline the design of the prompt predictor.

3.1.1. Prompt predictor network (prompt learning)

Given an image $I \in \mathbb{R}^{3 \times H \times W}$, the SAM image encoder extracts hierarchical feature embeddings $z = \{z_1, z_2, \dots, z_n\}$, where $z_i \in \mathbb{R}^{C_i \times H_i \times W_i}$ corresponds to the features extracted by a hierarchical transformer encoder ($n = 3$ in SAM2, and $n = 1$ otherwise). These feature maps capture multiscale representations, ranging from high-level semantic features in z_1 to low-level spatial features in z_n . We simply use the image embeddings from the last layer of the image encoder (i.e., z_1) as keys/values to the prompt prediction network. We explore the impact of utilizing these hierarchical features in the ablations 4.8.

The Prompt Predictor Network (PPN) uses learnable object queries $Q \in \mathbb{R}^{N \times C}$, where N is the number of queries. These queries are refined iteratively through n two-way cross-attention modules, which simultaneously update the queries and the image features at each level. The updated features at one level are added to the features at the next level, similar to how Feature Pyramid Networks (FPN) [47] constructs multiscale features by merging low-resolution features into high-resolution features before making predictions. Similarly, PPN begins with low-resolution semantic features and progressively refines them using higher-resolution spatial features.

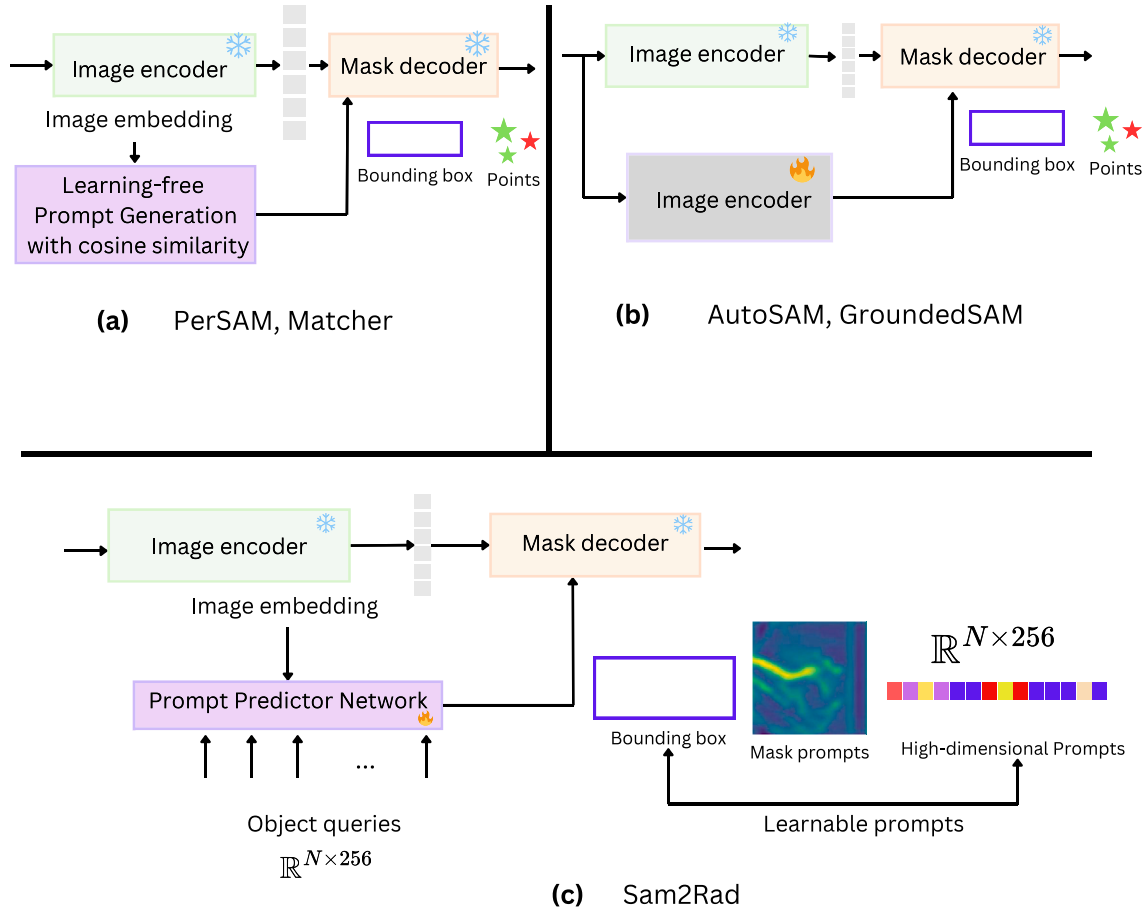


Fig. 3. Comparison between our prompt predictor network and others. PerSAM [29] and Matcher [30] use learning-free cosine similarity to predict box and point coordinates by comparing image patches between a reference image and a test image (a). Given image features, our *prompt predictor network* predicts prompts for the queried object. The predictor network outputs bounding box coordinates of target regions, an intermediate mask prompt, and N high-dimensional learnable embeddings ($\mathbb{R}^{N \times 256}$). All the predicted prompts (learnable prompts) are then fed to SAM's mask decoder to generate a mask. The prompt predictor network can handle multiple classes by providing learnable class queries (c). In contrast, Li et al. [31] and Xie et al. [32] predict only bounding boxes or intermediate masks, while GroundedSAM [28] and AutoSAM [33] predict prompts using a standalone image encoder (b).

Hierarchical attention mechanism. To incorporate spatial information, sinusoidal positional encodings $p_i \in \mathbb{R}^{C_i \times H_i \times W_i}$ are added to the feature embeddings z_i , yielding:

$$\tilde{z}_i = z_i + p_i.$$

At each level, the learnable object queries Q and $\tilde{z}_i$ are processed through a two-way cross-attention mechanism, resulting in updated queries $\hat{Q}$ and updated features $\hat{z}_i$:

$$\hat{Q}, \hat{z}_i = \text{TwoWayCrossAttention}(Q, \tilde{z}_i).$$

The updated features $\hat{z}_i$ are subsequently added to the features from lower level, with interpolation applied to match resolution and a 1×1 convolution to maintain the channel dimension:

$$\tilde{z}_{i+1} = z_{i+1} + \text{interpolate}(\hat{z}_i).$$

This approach parallels the Feature Pyramid Network (FPN) design, where lower-resolution feature maps enrich higher-resolution maps by contributing contextual information across scales. The process is iteratively applied for all hierarchical levels, as shown below:

$$\hat{Q}, \hat{z}_1 = \text{TwoWayCrossAttention}(Q, \tilde{z}_1),$$

$$\tilde{z}_2 = z_2 + \text{interpolate}(\hat{z}_1),$$

$$\hat{Q}, \hat{z}_2 = \text{TwoWayCrossAttention}(\hat{Q}, \tilde{z}_2),$$

$$\tilde{z}_3 = z_3 + \text{interpolate}(\hat{z}_2),$$

$$\dots,$$

$$Q^{(n)}, \hat{z}_n = \text{TwoWayCrossAttention}(Q^{(n-1)}, \tilde{z}_n).$$

The refined queries $Q^{(n)}$ capture object-specific information, while the multiscale refined image features $\hat{z}_n$ enhance the context for generating mask predictions. For SAM2, this process iterates through 3 hierarchical features maps ($n = 3$), while the original SAM uses a single level ($n = 1$)—first line of the computations above. Our experiments are based on single-level features, with hierarchical features explored in ablation studies (Section 4.8).

Prompt prediction. The final refined queries $Q^{(n)}$ are divided into components to generate the prompts:

- **Bounding Box Prediction:** The first two queries, $Q_{\text{box}} \in \mathbb{R}^{2 \times C}$, predict bounding box coordinates:

$$B = [x_{\min}, y_{\min}, x_{\max}, y_{\max}] = \text{Linear}(Q_{\text{box}}).$$

- **High-Dimensional Prompts:** The remaining $N - 2$ queries, $Q_{\text{hd}} \in \mathbb{R}^{(N-2) \times C}$, represent high-dimensional prompts.

Table 1

Number of images in the train/test sets.

Dataset	# of training images	# of test images	# of subjects
Hip	4987	4849	50
Wrist	3738	3822	101
Shoulder (Bursa)	7364	1605	206

- **Mask Prompts:** The refined image features $\hat{z}_n$ are passed through a convolutional block to generate a mask prompt:

$$\mathcal{M} = \text{Conv2D}(\hat{z}_n),$$

where $\mathcal{M} \in \mathbb{R}^{1 \times H/4 \times W/4}$ for an input image of dimensions $H \times W$.

The predicted bounding box coordinates ($\mathcal{B}$), high-dimensional prompts ($\mathcal{Q}_{\text{hd}}$), and mask prompts ($\mathcal{M}$) are then fed into SAM's mask decoder to produce the final segmentation mask.

Multi-class segmentation. For multi-class segmentation tasks, we simply introduce a set of learnable object queries Q_k specific to each class k . The architecture effectively scales to handle varying numbers of classes by simply extending the set of queries $Q = \{Q_1, Q_2, \dots, Q_C\}$, where C is the total number of classes. See Fig. 1 for model architecture.

3.2. Training losses

In some cases, the queried object may not be present in the image, such as in background images or when a specific class is missing in multi-class segmentation. To address this, we introduce a fully connected object prediction head and an output token within the mask decoder. During inference, if the object prediction head outputs a value less than 0, we disregard the segmentation output.

Given an image $I \in \mathbb{R}^{3 \times H \times W}$, a binary ground-truth mask $y \in \mathbb{R}^{H \times W}$, we obtain bounding box coordinates $\mathbb{R}^4$, mask prompts $\mathbb{R}^{1 \times 256 \times 256}$, and a predicted mask $\hat{y} \in \mathbb{R}^{H \times W}$ (after sigmoid activation), the training loss function is formulated as follows:

$$\mathcal{L}_{\text{Total}} = \lambda_1 \mathcal{L}_{\text{Mask}} + \lambda_2 \mathcal{L}_{\text{Box}} + \lambda_3 \mathcal{L}_{\text{Objectness}}, \quad (1)$$

where $\mathcal{L}_{\text{mask}}$ is the sum of focal loss [48] for the mask prompt, and the final mask prediction, where the focal loss is

$$\mathcal{L}_{\text{Focal}}(\hat{y}_i, y_i) = -y_i(1 - \hat{y}_i)^{\gamma} \log(\hat{y}_i) - \alpha(1 - y_i)\hat{y}_i^{\gamma} \log(1 - \hat{y}_i), \quad (2)$$

$\mathcal{L}_{\text{Box}}$ is a linear combination of L1 loss and the generalized IoU loss [49] for bounding box regression, and $\mathcal{L}_{\text{Objectness}}$ is the binary cross-entropy loss to predict the presence of the queried object in the image.

4. Experiments

4.1. Datasets

The datasets were curated from the US scans of different body parts including wrist, hip, and shoulder (Table 1) obtained with institutional ethics board approval. After splitting the subjects into training/test sets based on subject study ID, we extracted 2D frames from each volume and filtered out frames with no foreground objects.

4.2. Data augmentation

Overparametrized networks come with the risk of overfitting, especially when working with limited training data. To enhance the model's generalization and prevent overfitting, we apply strong augmentations: random horizontal/vertical flip, random 20% horizontal/vertical translations, a random rotation in the range $[-90, 90]$ degrees, and a random resized crop in the range of $[0.8, 1.0]$. Each augmentation is applied with a probability of 0.5. Fig. 4 illustrates an example of such augmentation and demonstrates the model's robustness to these augmentations.

Table 2

Zero-shot Dice/IoU scores of SAM variants on three ultrasound datasets. Models were prompted with ground truth bounding boxes.

Model	Hip	Wrist	Shoulder
MobileSAM [24]	68.2/54.1	50.8/35.9	18.1/10.2
SAM ViTB	76.7/62.9	59.3/43.7	22.2/12.9
MedSAM [25]	14.5/8.3	20.5/11.8	13.1/7.2
SAM ViTH	75.4/62.5	54.7/39.8	18.5/10.6
SAM2 (Hiera-Tiny)	75.1/61.2	49.9/35.3	15.3/8.8
SAM2 (Hiera-Small)	74.7/60.9	45.6/31.9	13.0/7.6
SAM2 (Hiera-Large)	75.2/61.0	59.5/44.3	25.2/15.1
SAM2 (Hiera-Base+)	78.2/65.2	63.5/47.9	25.3/15.1

Table 3

Comparison of bounding box and learned prompts on hip and wrist datasets using a frozen SAM2. The results demonstrate that learned prompts perform better than bounding box prompts without fine-tuning any SAM2 parameters. Our results are highlighted in gray. Scores are presented as Dice/IoU.

Model	Bounding Box		Learned Prompts	
	Hip	Wrist	Hip	Wrist
SAM2 (Hiera-Tiny)	75.1/61.2	49.9/35.3	81.2/69.0	82.7/71.1
SAM2 (Hiera-Small)	74.7/60.9	45.6/31.9	80.5/68.1	83.1/71.6
SAM2 (Hiera-Large)	75.2/61.0	59.5/44.3	81.6/70.0	83.1/71.7
SAM2 (Hiera-Base+)	78.2/65.2	63.5/47.9	80.4/68.0	83.1/71.7

4.3. Implementation details

The code is implemented using PyTorch [50]. We used the AdamW optimizer [51] with a weight decay of 0.1, batch size of 4, learning rate of $1e-4$ for the prompt predictor network, and $1e-5$ for the mask decoder parameters. $\lambda_1, \lambda_2, \lambda_3$ in Eq. (1) are set to 10, 1, 1, respectively. The focal loss has parameters $\gamma = 3$ and $\alpha = 0.7$ in Eq. (2). To maintain the aspect ratio, we resized the longer side of each image to 1024 pixels and padded the shorter side with zeros.

4.4. Evaluation metrics

Models' performance was evaluated using Intersection over Union (IoU) and the Dice similarity coefficient on a held-out test set. Table 1 shows the number of test images for each dataset used in this study.

4.5. Does prompt learning work any better than bounding box prompts?

To evaluate the effectiveness of learned prompts versus bounding box prompts, we conducted an experiment using frozen SAM2. The results, presented in Table 3, demonstrate that learned prompts consistently outperform bounding box prompts across all SAM2 variants. For instance, on the wrist dataset, SAM2 (Hiera-Base+) achieved a Dice score of 83.5% with learned prompts, compared to 63.5% with bounding box prompts. These findings suggest that the prompt learning strategy not only addresses SAM's need for manual prompts but also introduces additional information not present in manual bounding box prompts. The full results, including LoRA [42] fine-tuning the mask decoder, are shown in Table 4.

4.6. Zero-shot and few-shot segmentation

First, we evaluate the zero-shot generalizability of frozen SAM, SAM2, and MedSAM [25] (fine-tuned on several medical imaging modalities) on three US datasets using ground truth bounding boxes as prompts. The results in Table 2 show that SAM2 outperforms SAM on all datasets, with the most significant improvement observed on the wrist dataset. These findings demonstrate that SAM2 aligns better with US images, even without fine-tuning. However, on more challenging datasets such as the shoulder dataset, both SAM and SAM2 perform

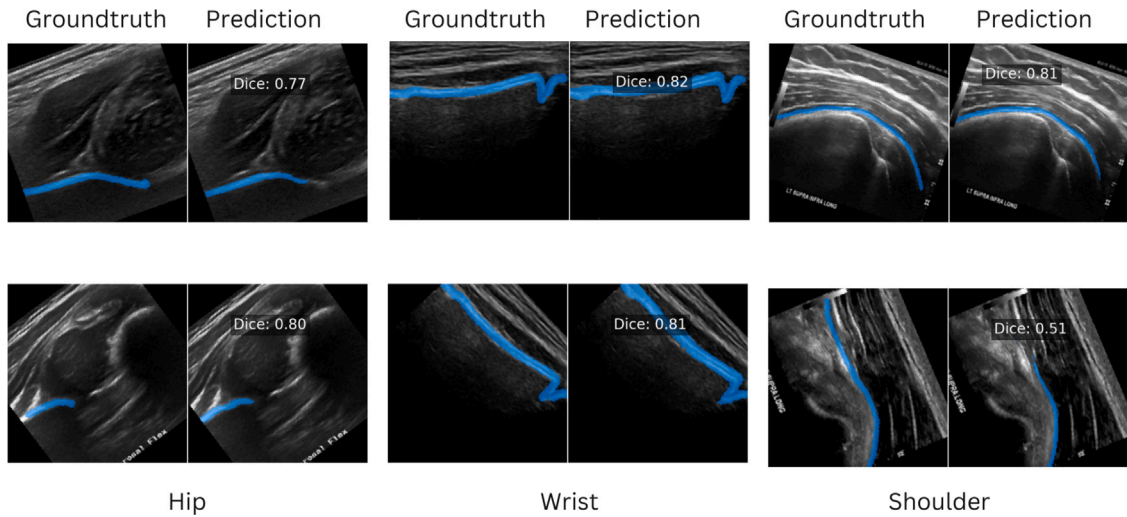


Fig. 4. The model is robust to strong augmentations. For details on the data augmentation techniques applied during training, see Section 4.2.

Table 4

Sam2Rad Dice/IoU results. LP - Learned Prompts, BB + LP - Bounding Box plus Learned Prompts.

Model	Hip		Wrist		Shoulder (bursa)	
	LP	BB+LP	LP	BB+LP	LP	BB+LP
MobileSAM	78.4/65.7	81.4/69.2	79.8/67.3	81.4/69.3	73.7/60.7	75.0/62.0
SAM-ViT	81.9/70.3	84.6/73.8	82.6/71.0	83.9/72.8	75.4/62.9	76.0/63.5
SAM2Rad (Hiera-Tiny)	82.4/70.9	84.0/73.0	83.6/72.4	84.6/73.8	76.3/63.9	77.6/65.3
SAM2Rad (Hiera-Small)	82.6/71.1	84.7/73.9	84.5/73.7	85.5/75.1	74.8/61.9	75.6/62.7
SAM2Rad (Hiera-Large)	81.6/69.9	84.3/73.5	76.5/64.1	77.2/64.5	76.5/64.1	77.2/64.7
SAM2Rad (Hiera-Base+)	82.9/71.7	85.5/75.2	83.7/72.6	85.1/75.2	75.0/62.1	75.6/62.7

Table 5

Dice scores for different prompt configurations.

Prompts	Dice score (%)
Bounding Box (BB)	45.1
High-Dimensional Prompts (HDP)	79.8
BB + Mask	81.0
BB + HDP	80.2
Mask + HDP	82.6
BB + Mask + HDP	83.3
BB + Mask + HDP + Hiera	83.7
Non-SAM models	
UNet [1]	78.72
UNet++ [4]	80.05

poorly, indicating the need for fine-tuning. Surprisingly, MedSAM, despite being fine-tuned on medical images, underperforms compared to SAM and SAM2 on US images, even though the latter two were trained solely on natural images.

To further validate the effectiveness of our approach in limited-data settings, we trained the model using only 10 labeled images randomly sampled from the hip training set and evaluated it on the full test set. Although the model could potentially benefit from retrieving hard or more representative examples, we selected subsets randomly and used the same set images to train all models. As illustrated in Fig. 5, the proposed prompting strategy consistently performs well across all SAM variants, with SAM2 achieving the highest accuracy. These results align with our findings from the ablation studies conducted on the wrist dataset (Table 5), highlighting Sam2Rad's potential for medical imaging applications where expert annotations may be scarce.

4.7. Robustness to noisy ground truth masks

Fig. 4 demonstrates the models' robustness to heavy data augmentation on the test set. This is crucial for generalization to unseen data,

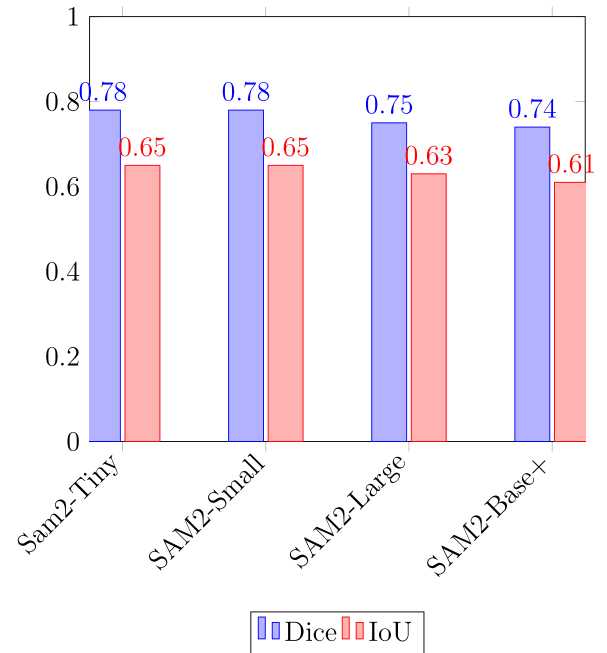


Fig. 5. Prompting SAM2 with few-shot images. The figure shows the performance of various SAM variants trained on just 10 images from the hip dataset and evaluated on the entire test dataset. The proposed prompting strategy demonstrates data efficiency across all SAM variants.

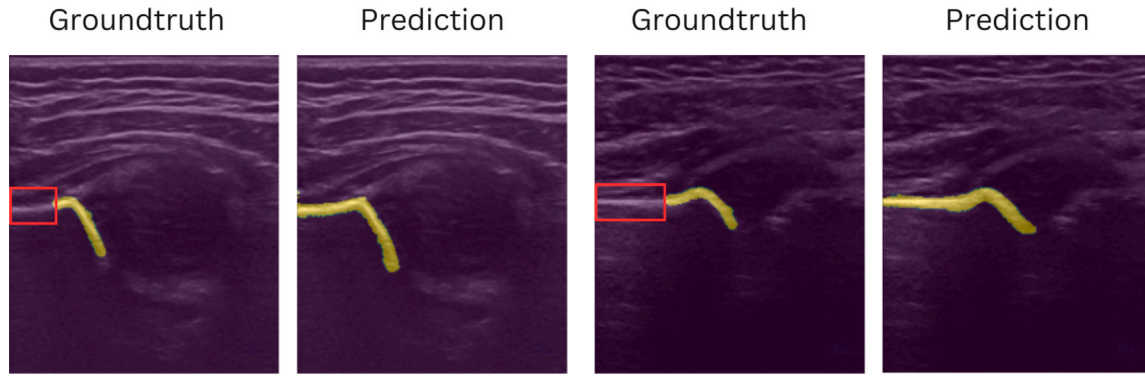


Fig. 6. Sam2-Tiny with PPN predictions on test images. The model provides complete bone segmentation, in contrast to the incomplete ground truth labels.

especially in overparameterized models like many foundation models. The results suggest that the model is learning the underlying structure of the data rather than simply memorizing the training set. Our labels contained various inconsistencies, for instance, partial masks. Despite these imperfections, the model effectively averaged out the noise, producing accurate segmentations on test images. Fig. 6 provides visual examples of this robustness.

4.8. Ablation studies

To evaluate the contributions of individual components in the Sam2Rad framework, we conducted a series of ablation studies. These experiments focused on evaluating different prompt configurations and the impact of hierarchical feature maps while keeping all SAM2 parameters frozen. This setup isolates the effects of the Prompt Predictor Network (PPN) and ensures that observed improvements are due to the proposed prompt-learning methodology.

Experimental setup. We used the smallest version of SAM2 (SAM2-Tiny) for these experiments and froze all its parameters to clearly attribute performance improvements to the learned prompts. The evaluation was conducted under a few-shot setting, using 10 randomly selected labeled images from the wrist dataset. This setup ensures a controlled comparison of bounding box, mask, high-dimensional embeddings, and combined prompts. Finally, we evaluated the impact of using hierarchical features extracted from SAM2's image encoder to incorporate multiscale information. The possible prompt configurations are listed below:

1. **Bounding Box (BB):** Bounding box coordinates predicted by the PPN were used as prompts.
2. **Mask (M):** The mask prompt cannot be used in isolation since it relies on randomly initialized queries that are refined using a two-way attention mechanism (as shown in Fig. 1). The mask prompt must be used in combination with object queries, which simultaneously update both the queries and the mask during the prompt generation.
3. **High-Dimensional Prompts (HDP):** $N \times 256$ -d embeddings predicted by PPN are used as prompts. These prompts are a result of the two-way cross-attention mechanism between the object queries and the image embeddings.
4. **Combined Prompts:** Includes Box + Mask, Box + HDP, Mask + HDP, and Box + Mask + HDP.
5. **Box + Mask + HDP + Hierarchical Features (+Hiera):** Multiscale feature maps were incorporated to enhance the PPN's prompt prediction process.

Results and analysis. The results are summarized in Table 5. Using only bounding box prompts achieved a Dice score of 45.1%, slightly lower than the 49.9% obtained with ground truth boxes (Table 3). This indicates that while PPN can reliably predict bounding box coordinates, additional contextual information is beneficial. The combination of mask prompts with high-dimensional embeddings yielded substantial improvements, achieving Dice scores above 82%, and the combined prompts achieved a Dice score of 83.3%. This demonstrates that high-dimensional embeddings effectively capture complementary contextual information beyond what geometric prompts alone provide.

Including hierarchical features (+Hiera) showed only marginal improvements in segmentation performance, with a marginal 0.4%–0.6% increase in Dice score across configurations. This small improvement suggests that while multiscale spatial and semantic features can contribute to the prompt prediction process, their impact is not substantial for the current datasets. The more significant gains came from combining different prompt types, indicating that the complementary information from various prompt modalities is more crucial. The potential benefits of hierarchical features could be further investigated for small object segmentation tasks in future work, where fine-grained multiscale information may prove more valuable.

These ablation studies reveal that: (1) combined prompts consistently outperform individual configurations, (2) high-dimensional embeddings provide crucial contextual information beyond geometric prompts, and (3) hierarchical features enhance prompt prediction accuracy. Notably, these improvements were achieved in a few-shot setting with only 10 training images, demonstrating the efficiency of our prompting strategy even with limited data.

Comparison to non-SAM models. SAM2Rad achieves comparable performance to specialized models like UNet [1] and UNet++ [4] while providing greater flexibility and generalization (Table 5). By leveraging the SAM foundation model, SAM2Rad benefits from inherent robustness to diverse datasets and scenarios. The Prompt Predictor Network (PPN) seamlessly integrates with SAM or similar architectures, ensuring adaptability to future advancements.

Additionally, SAM2Rad supports an interactive mode, enabling users to refine segmentations through additional prompts—an advantage over static specialized models. Its ability to combine automated and semi-automated workflows makes it a versatile tool for clinical applications.

4.9. Applications

Medical image segmentation is essential for clinical analysis, enabling precise delineation of anatomical structures for diagnosis and quantitative measurements. In ultrasound imaging, accurate segmentation is particularly valuable due to the modality's real-time capabilities, cost-effectiveness, and non-invasive nature. For our purposes, segmentation helps isolate bony structures and potential fracture

sites, enabling automated detection of abnormalities. We demonstrate Sam2Rad's effectiveness on three musculoskeletal ultrasound datasets targeting wrist, hip, and shoulder regions. These anatomical sites were chosen because they represent common areas for fracture assessment and often require precise segmentation for clinical evaluation.

4.10. Future work

Future work on Sam2Rad includes extending the framework to additional imaging modalities, such as CT, MRI, and X-ray, to broaden its applicability. Adapting the model to handle 3D medical images, such as volumetric CT or MRI data, would significantly enhance its utility, requiring modifications like 3D convolutional layers or volumetric prompt prediction mechanisms. Currently, Sam2Rad is designed for 2D images only. Furthermore, distilling the trained model into smaller, task-specific versions could enable real-time applications in clinical environments without compromising accuracy. These advancements would further enhance Sam2Rad's usability, scalability, and integration into diverse clinical workflows, establishing it as a practical tool for medical image segmentation.

5. Conclusion

In this paper, we presented Sam2Rad, a novel prompt learning approach that significantly enhances the performance of SAM/SAM2 and its variants for ultrasound image segmentation. Our method addresses key limitations of promptable models by eliminating the need for manual prompts and improving segmentation accuracy across various datasets. Sam2Rad outperformed all SAM and SAM2 variants in zero-shot generalization, particularly for challenging cases like shoulder ultrasound. The model demonstrated remarkable improvements, with Dice score increases ranging from a 2.2%–5.8% for hip, 19.6%–32.8% for wrist, and up to 51.3% improvement in Dice score (from 25.2% to 76.5% Sam2 large) on shoulder data. Notably, Sam2Rad can be trained with as few as 10 labeled images. By leveraging SAM's image encoder features, our network seamlessly integrates with SAM's architecture to produce high-quality prompts. Our model offers three modes: autonomous, semi-autonomous with human-in-the-loop, and fully manual. The seamless integration with SAM's architecture and compatibility with any SAM variant make Sam2Rad a versatile and powerful tool for automatic segmentation in medical imaging.

CRediT authorship contribution statement

Assefa Seyoum Wahd: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Banafshe Felfeliyan:** Writing – review & editing, Validation. **Yuyue Zhou:** Writing – review & editing, Validation, Data curation. **Shrimanti Ghosh:** Writing – review & editing, Data curation. **Adam McArthur:** Writing – review & editing, Data curation. **Jiechen Zhang:** Writing – review & editing. **Jacob L. Jaremko:** Writing – review & editing, Supervision, Resources, Funding acquisition, Data curation, Conceptualization. **Abhilash Hareendranathan:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] O. Ronneberger, P. Fischer, T. Brox, U-net: convolutional networks for biomedical image segmentation, 2015, <https://arxiv.org/abs/1505.04597v1>.
- [2] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask r-CNN, 2018.
- [3] F. Isensee, J. Petersen, A. Klein, D. Zimmerer, P.F. Jaeger, S. Kohl, J. Wasserthal, G. Koehler, T. Norajitra, S. Wirkert, K.H. Maier-Hein, 'Nnu-net: self-adapting framework for u-net-based medical image segmentation, 2018.
- [4] Z. Zhou, M.M.R. Siddiquee, N. Tajbakhsh, J. Liang, Unet++: a nested U-net architecture for medical image segmentation, 2018.
- [5] O. Oktay, J. Schlemper, L.L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N.Y. Hammerla, B. Kainz, B. Glocker, D. Rueckert, Attention u-net: learning where to look for the pancreas, 2018.
- [6] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A.L. Yuille, Y. Zhou, TransUNet: transformers make strong encoders for medical image segmentation, 2021.
- [7] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H. Roth, D. Xu, UNETR: transformers for 3D medical image segmentation, 2021.
- [8] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, M. Wang, Swin-unet: unet-like pure transformer for medical image segmentation, 2021.
- [9] J. Kaplan, S. McCandlish, T. Henighan, T.B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, D. Amodei, Scaling laws for neural language models, 2020.
- [10] J.S. Rosenfeld, A. Rosenfeld, Y. Belinkov, N. Shavit, A constructive prediction of the generalization error across scales, 2019.
- [11] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, 2021.
- [12] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A.C. Berg, W.-Y. Lo, P. Dollár, R. Girshick, Segment anything, 2023.
- [13] T.B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D.M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, D. Amodei, Language models are few-shot learners, 2020.
- [14] M. Käppler, K. Petek, N.V. Vödisch, W. Burgard, A. Valada, Few-shot panoptic segmentation with foundation models, in: 2024 IEEE International Conference on Robotics and Automation, ICRA, 2024, pp. 7718–7724.
- [15] T. Chen, Z. Mai, R. Li, W. I. Chao, Segment anything model (SAM) enhanced pseudo labels for weakly supervised semantic segmentation, 2023.
- [16] C. Schlarmann, M. Hein, On the adversarial robustness of multi-modal foundation models, 2023.
- [17] A. Zhou, J. Wang, Y.-X. Wang, H. Wang, Distilling out-of-distribution robustness from vision-language foundation models, in: Thirty-Seventh Conference on Neural Information Processing Systems, 2023.
- [18] Y. Ge, J. Ren, A. Gallagher, Y. Wang, M.-H. Yang, H. Adam, L. Itti, B. Lakshminarayanan, J. Zhao, Improving zero-shot generalization and robustness of multi-modal models, 2023.
- [19] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryal, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, C. Feichtenhofer, SAM 2: Segment Anything in Images and Videos.
- [20] 'LR0.FM: LOW-resolution ZERO-shot classification BENCHMARK FOR foundation MODELS, in: The Thirteenth International Conference on Learning Representations, 2024.
- [21] C. Wang, R. Jia, X. Liu, D. Song, Benchmarking zero-shot robustness of multimodal foundation models: a pilot study, 2024.
- [22] M.A. Mazurowski, H. Dong, H. Gu, J. Yang, N. Konz, Y. Zhang, Segment anything model for medical image analysis: An experimental study, in: Medical Image Analysis, Vol. 89, 2023, 102918.
- [23] C. Mattjie, L.V. de Moura, R.C. Ravazio, L.S. Kupssinskü, O. Parraga, M.M. Delucis, R.C. Barros, Zero-shot performance of the segment anything model (SAM) in 2D medical imaging: A comprehensive evaluation and practical guidelines, 2023.
- [24] C. Zhang, D. Han, Y. Qiao, J.U. Kim, S.-H. Bae, S. Lee, C.S. Hong, Faster segment anything: towards lightweight SAM for mobile applications, 2023.
- [25] J. Ma, Y. He, F. Li, L. Han, C. You, B. Wang, Segment anything in medical images, 2023.

- [26] T. Chen, L. Zhu, C. Ding, R. Cao, Y. Wang, Z. Li, L. Sun, P. Mao, Y. Zang, SAM fails to segment anything? – SAM-adapter: adapting SAM in underperformed scenes: camouflage, shadow, medical image segmentation, and more, 2023, <https://arxiv.org/abs/2304.09148v3>.
- [27] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, C. Li, J. Yang, H. Su, J. Zhu, L. Zhang, Grounding DINO: marrying DINO with grounded pre-training for open-set object detection, 2023, <https://arxiv.org/abs/2303.05499v4>.
- [28] T. Ren, S. Liu, A. Zeng, J. Lin, K. Li, H. Cao, J. Chen, X. Huang, Y. Chen, F. Yan, Z. Zeng, H. Zhang, F. Li, J. Yang, H. Li, Q. Jiang, L. Zhang, Grounded SAM: assembling open-world models for diverse visual tasks, 2024.
- [29] R. Zhang, Z. Jiang, Z. Guo, S. Yan, J. Pan, X. Ma, H. Dong, P. Gao, H. Li, Personalize segment anything model with one shot, 2023.
- [30] Y. Liu, M. Zhu, H. Li, H. Chen, X. Wang, C. Shen, Matcher: segment anything with one shot using all-purpose feature matching, 2023, <https://arxiv.org/abs/2305.13310v1>.
- [31] J. Li, Z. Zhang, H. Zhao, Self-prompting large language models for zero-shot open-domain QA, 2023.
- [32] B. Xie, H. Tang, B. Duan, D. Cai, Y. Yan, MaskSAM: towards auto-prompt SAM with mask classification for medical image segmentation, 2024.
- [33] T. Shaharabany, A. Dahan, R. Giryas, L. Wolf, Autosam: adapting SAM to medical images by overloading the prompt encoder, 2023.
- [34] P. Shi, J. Qiu, S.M.D. Abaxi, H. Wei, F.P.-W. Lo, W. Yuan, Generalist vision foundation models for medical imaging: a case study of segment anything model on zero-shot medical segmentation, in: *Diagnostics*, Vol. 13, 2023, p. 1947, 11.
- [35] S. Roy, T. Wald, G. Koehler, M.R. Rokuss, N. Disch, J. Holzschuh, D. Zimmerer, K.H. Maier-Hein, SAM.MD: zero-shot medical image segmentation capabilities of the segment anything model, 2023.
- [36] Y. Huang, X. Yang, L. Liu, H. Zhou, A. Chang, X. Zhou, R. Chen, J. Yu, J. Chen, C. Chen, S. Liu, H. Chi, X. Hu, K. Yue, L. Li, V. Grau, D.-P. Fan, F. Dong, D. Ni, Segment anything model for medical images?, in: *Medical Image Analysis*, vol. 92, 2024, 103061.
- [37] J. Cheng, J. Ye, Z. Deng, J. Chen, T. Li, H. Wang, Y. Su, Z. Huang, J. Chen, L. Jiang, H. Sun, J. He, S. Zhang, M. Zhu, Y. Qiao, SAM-Med2D, 2023.
- [38] Q. Wu, Y. Zhang, M. Elbatel, Self-prompting large vision models for few-shot medical image segmentation, in: L. Koch, M.J. Cardoso, E. Ferrante, K. Kamnitsas, M. Islam, M. Jiang, N. Rieke, S.A. Tsaftaris, D. Yang (Eds.), *Domain Adaptation and Representation Transfer*, vol. 14293, Springer Nature Switzerland, Cham, 2024, pp. 156–167.
- [39] W. Lei, X. Wei, X. Zhang, K. Li, S. Zhang, ‘Medlsam: localize and segment anything model for 3D medical images, 2023.
- [40] K. Zhou, J. Yang, C.C. Loy, Z. Liu, K. Zhou, J. Yang, C.C. Loy, Z. Liu, Conditional prompt learning for vision-language models,, 2022.
- [41] J. Wu, Y. Zhang, R. Fu, H. Fang, Y. Liu, Z. Wang, Y. Xu, Y. Jin, Medical SAM adapter: adapting segment anything model for medical image segmentation, 2023.
- [42] E.J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, Lora: low-rank adaptation of large language models, 2021.
- [43] K. Zhang, D. Liu, Customized segment anything model for medical image segmentation,, 2023, arXiv.org.
- [44] H. Gu, H. Dong, J. Yang, M.A. Mazurowski, How to build the best medical image segmentation algorithm using foundation models: A comprehensive empirical study with segment anything model, 2024.
- [45] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An image is worth 16x16 words: transformers for image recognition at scale, 2020, <https://arxiv.org/abs/2010.11929v2>.
- [46] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, Masked autoencoders are scalable vision learners, 2021, <https://arxiv.org/abs/2111.06377v3>.
- [47] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature pyramid networks for object detection, 2017.
- [48] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, 2018.
- [49] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, S. Savarese, Generalized intersection over union: a metric and a loss for bounding box regression, 2019.
- [50] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimeshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, S. Chintala, Pytorch: an imperative style, high-performance deep learning library, 2019.
- [51] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, 2019.